

INTRODUCTION TO PROBABILITY AND STATISTICS FOURTEENTH EDITION

Chapter 8 Large-Sample Estimation

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INTRODUCTION

- Populations are described by their probability distributions and parameters.
 - For quantitative populations, the location and shape are described by μ and σ .
 - For a binomial populations, the location and shape are determined by p .
- If the values of parameters are unknown, we make inferences about them using sample information.

A decorative graphic on the left side of the slide consists of several vertical stripes in various shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. One circle contains the number '3'.

8.1 WHERE WE'VE BEEN AND WHERE WE'RE GOING

3

STATISTICAL INFERENCE

- **Estimation:**

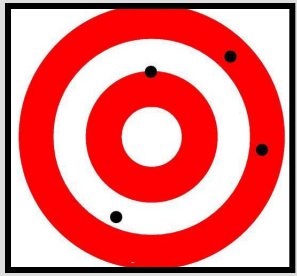
- **Point estimation:** Find an estimator
估計量.
- **Interval estimation:** Find a
confidence interval

- **Hypothesis Testing:**

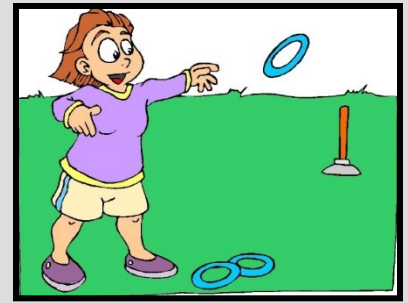
- Deciding about the value of a
parameter based on some
preconceived idea.

A STATISTIC: A STANDARDIZED ESTIMATOR

- **Interval estimation:** Find a confidence interval
- **Hypothesis Testing:**
 - Deciding about the value of a parameter based on some preconceived idea.



ESTIMATOR



- An estimator is a rule, usually a formula, that tells you how to calculate the estimate based on the sample.
- Since an estimator is calculated from sample values, it varies from sample to sample according to its **sampling distribution**.
- An **estimator** is **unbiased** if the mean of its sampling distribution equals the parameter of interest.

EXAMPLES



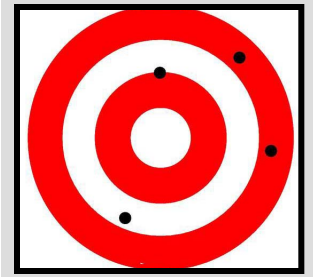
- A consumer wants to estimate the price of similar homes in her city before putting her home on the market.

Estimation: Estimate μ , the average home price.

- A manufacturer wants to know if a new type of steel is more resistant to high temperatures than an old type was.

Hypothesis test: Is the new average resistance, μ_N equal to the old average resistance, μ_O ?

NOTATIONS



- Before sampling, we define an *estimator* 估計量, is a random variable.
- We use *uppercase* in our slides.
- After sampling, we obtain an *estimate* 估計值, which is a realized value of the estimator, is a number.
- We use *lowercase* in our slides
- This textbook does not distinguish these two different concepts...

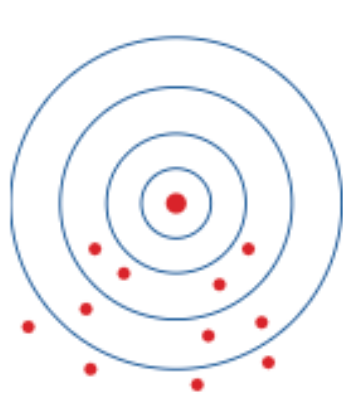
A decorative vertical stripe on the left side of the slide, composed of several thin, parallel lines in various shades of blue. To the right of this stripe, there are several overlapping circles of different sizes, also in shades of blue. One circle is particularly large and prominent, while others are smaller and more subtle.

8.2 POINT ESTIMATION, ONE POPULATION PROPORTION

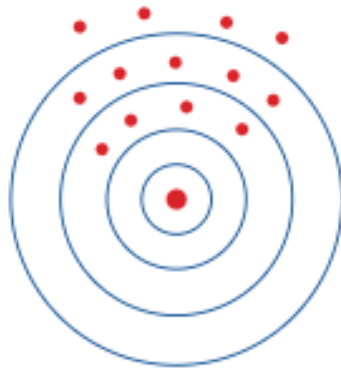


Figure 8.1

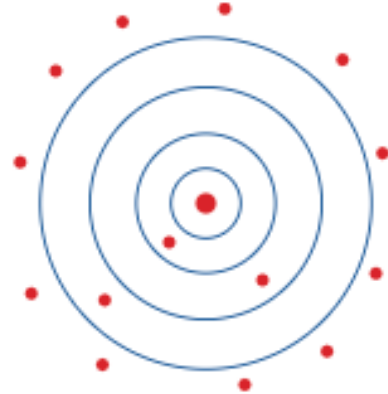
Which marksman is best?



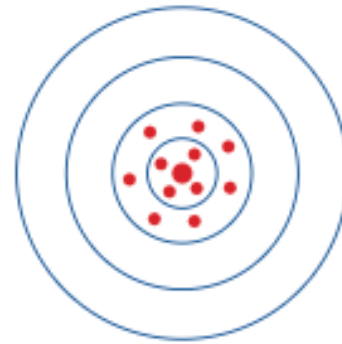
Mostly below
bull's-eye



Mostly above
bull's-eye



Off bull's-eye
by a wide margin



Best
marksmanship

Figure 8.2

Distributions for biased and unbiased estimators

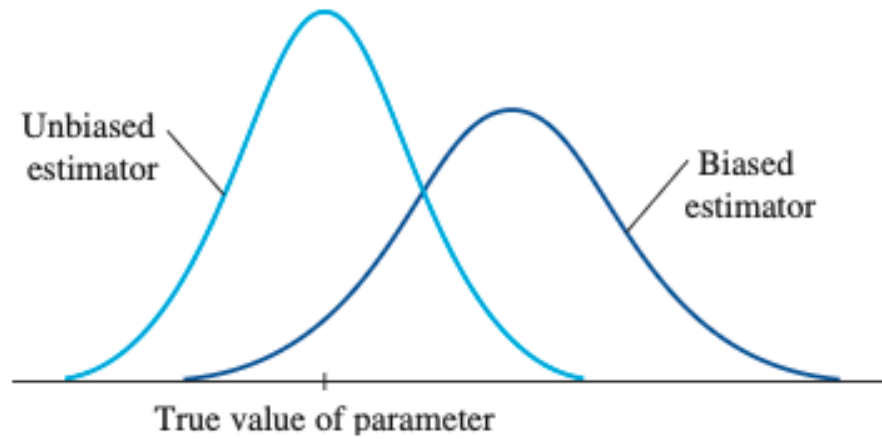
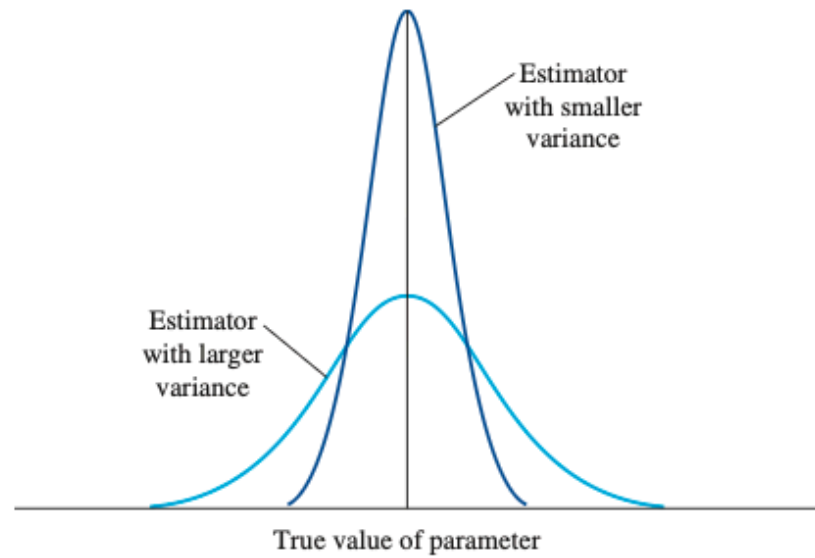
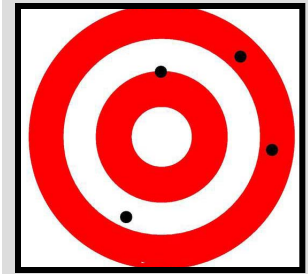


Figure 8.3

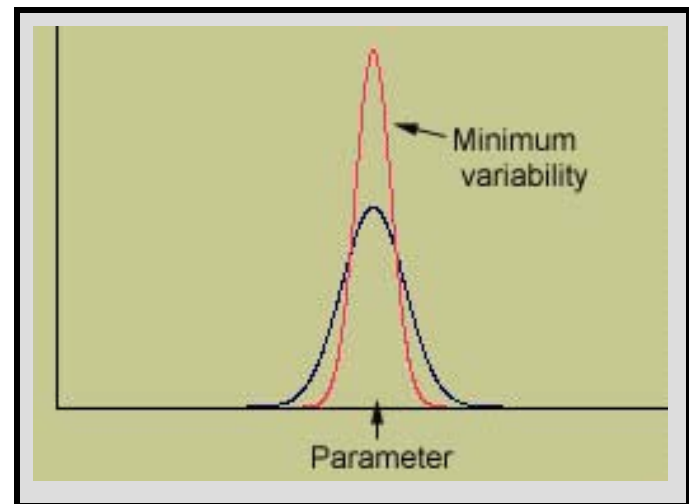
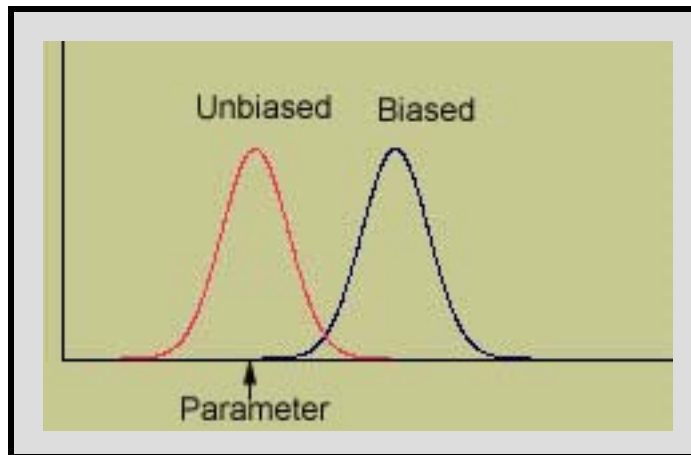
Comparison of estimator
variability



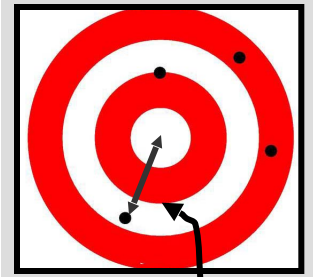
PROPERTIES OF POINT ESTIMATORS



- Of all the **unbiased** estimators, we prefer the estimator whose sampling distribution has the **smallest spread** or **variability**.



MEASURING THE GOODNESS OF AN ESTIMATOR



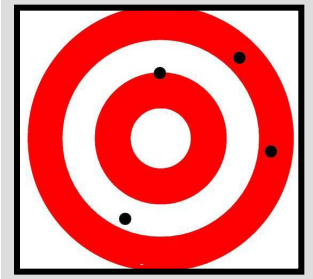
- The distance between an estimate and the true value of the parameter is the **error of estimation**.

The distance between the bullet and the bull's-eye.

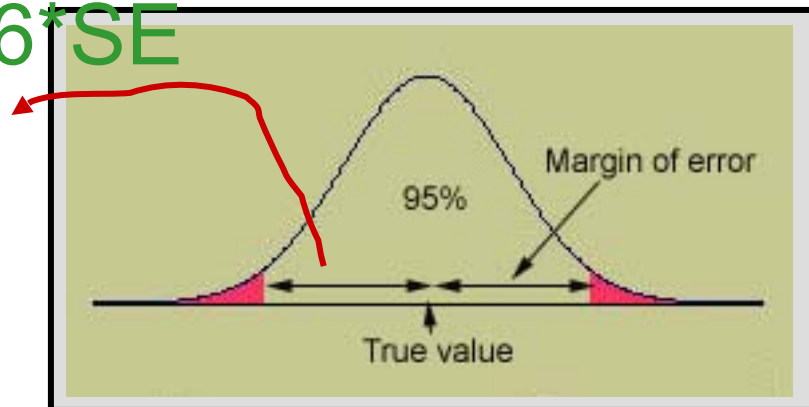
- In this chapter, the sample sizes are large, so that our ***unbiased*** estimators will have **normal** distributions.

Because of the Central Limit Theorem.

THE MARGIN OF ERROR



- For *unbiased* estimators with normal sampling distributions, 95% of all point estimates will lie within 1.96 standard deviations of the parameter of interest.
- **95% Margin of error (95% 誤差範圍)**
- The maximum error of estimation, calculated as $1.96 * SE$



The slide features a dark blue background. On the left side, there are several vertical stripes of varying shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. The largest circle is positioned near the top left, and several smaller circles are arranged vertically below it.

8.3 INTERVAL ESTIMATION

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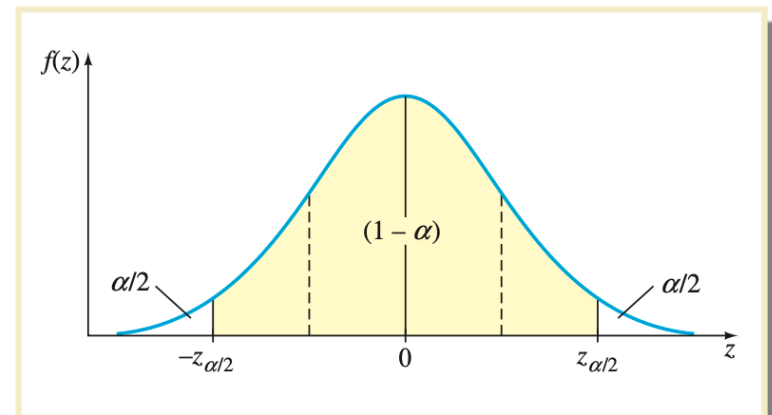
THE $100(1 - \alpha)\%$ CONFIDENCE INTERVAL (CI) FOR μ .

- Let $X_i \stackrel{i.i.d.}{\sim} \mathcal{G}(\mu, \sigma^2)$ for $i = 1, \dots, n$.
- By CLT, as $n \rightarrow \infty$:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1).$$

- Consider the probability

$$\mathbb{P}\left(-z_{\alpha/2} \leq \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha.$$



- Rearranging the inequality, we have

$$\mathbb{P} \left(-z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \bar{X}_n - \mu \leq z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) = 1 - \alpha.$$

- $$\mathbb{P} \left(-\bar{X}_n - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq -\mu \leq -\bar{X}_n + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) = 1 - \alpha.$$

- Further simplifying, we obtain

$$\mathbb{P} \left(\bar{X}_n - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X}_n + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) = 1 - \alpha.$$

- Thus, the $(1 - \alpha) \%$ confidence interval for μ is given by:

- $$\left[\bar{X}_n - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right], \text{ or, simply, } \bar{X}_n \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

How do we interpret this CI for μ ?

- Further simplifying, we obtain

$$\mathbb{P} \left(\bar{X}_n - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X}_n + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) = 1 - \alpha.$$

In repeated sampling, about $100(1 - \alpha)\%$ intervals constructed this way will cover the true parameter μ .

Textbook p. 301

● Need a Tip?

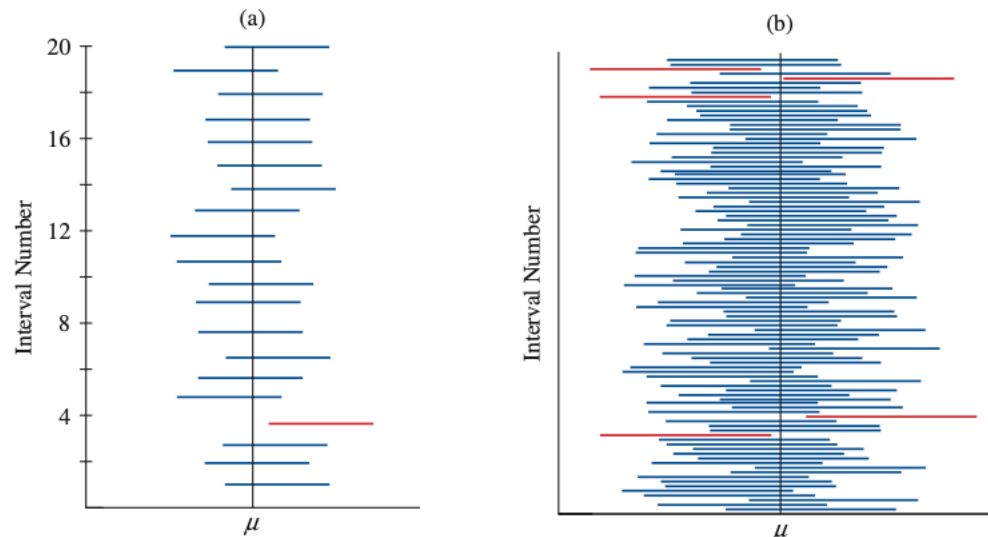
A 95% confidence interval tells you that, if you were to construct many of these intervals (all of which would have slightly different endpoints), 95% of them would enclose the population mean.

Figure 8.9

Interpreting confidence intervals

■ Interpreting the Confidence Interval

What does it mean to say you are “95% confident” that the true value of the population mean μ is within a given interval? If you were to construct 20 such intervals, each using different sample information, your intervals might look like those shown in Figure 8.9(a).



How do we interpret this CI?
In repeated sampling, about $100(1 - \alpha) \%$ intervals constructed this way will converge to the true parameter.

THE $100(1 - \alpha) \%$ MARGIN OF ERROR

Table 3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011
-2.9	.0019	.0018	.0017	.0017	.0016	.0016	.0015
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594

- The CI: $\bar{X}_n \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

- The $100(1 - \alpha) \%$ margin of error is $z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

- 95% Margin of error (95% 誤差範圍) is $1.96 \frac{\sigma}{\sqrt{n}}$

- $\frac{\sigma}{\sqrt{n}}$ is the standard error (SE) of the estimator $\bar{X}_n$

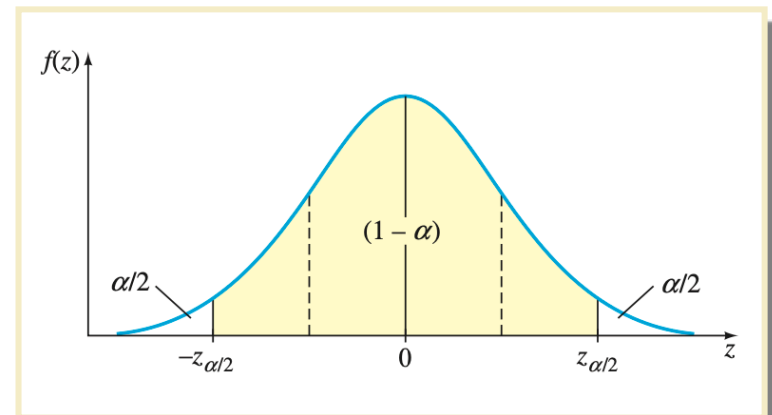
IF σ IS UNKNOWN, WE ESTIMATE IT BY S

- Let $X_i \stackrel{i.i.d.}{\sim} \mathcal{G}(\mu, \sigma^2)$.
- The Slutsky Theorem (CLT), as $n \rightarrow \infty$:

$$\frac{\bar{X}_n - \mu}{S/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1).$$

- Consider the probability

$$\mathbb{P} \left(-z_{\alpha/2} \leq \frac{\bar{X}_n - \mu}{S/\sqrt{n}} \leq z_{\alpha/2} \right) = 1 - \alpha.$$



THE $100(1 - \alpha)\%$ CI FOR μ .

- Rearranging the inequality, we have

$$\mathbb{P} \left(-z_{\alpha/2} \frac{S}{\sqrt{n}} \leq \bar{X}_n - \mu \leq z_{\alpha/2} \frac{S}{\sqrt{n}} \right) = 1 - \alpha.$$

- Further simplifying, we obtain

$$\mathbb{P} \left(\bar{X}_n - z_{\alpha/2} \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X}_n + z_{\alpha/2} \frac{S}{\sqrt{n}} \right) = 1 - \alpha.$$

- Thus, the $(1 - \alpha)\%$ confidence interval for μ is given by:

- $\left[\bar{X}_n - z_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X}_n + z_{\alpha/2} \frac{S}{\sqrt{n}} \right]$, or, simply, $\bar{X}_n \pm z_{\alpha/2} \frac{S}{\sqrt{n}}$

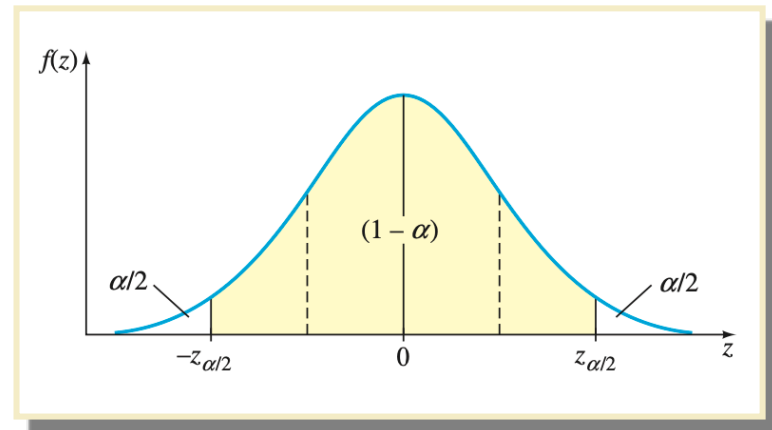
IF p IS UNKNOWN, WE ESTIMATE IT BY $\hat{p}$

- Let $X_i \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$. Then $\bar{X}_n = \hat{p}$ and $\hat{q} = 1 - \hat{p}$.
- The Slutsky Theorem (CLT), as $n \rightarrow \infty$:

$$\frac{\hat{p} - p}{\sqrt{\hat{p}\hat{q}/n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

- Consider the probability

$$\mathbb{P}\left(-z_{\alpha/2} \leq \frac{\hat{p} - p}{\sqrt{\hat{p}\hat{q}/n}} \leq z_{\alpha/2}\right) = 1 - \alpha.$$



THE $100(1 - \alpha) \%$ CI FOR p IN $\text{BERNOULLI}(p)$.

- Start from the probability (with $\hat{q} = 1 - \hat{p}$):

$$\mathbb{P}\left(-z_{\alpha/2} \leq \frac{\hat{p} - p}{\sqrt{\hat{p}\hat{q}/n}} \leq z_{\alpha/2}\right) = 1 - \alpha.$$

- Rearranging the inequality, we have

$$\mathbb{P}\left(-z_{\alpha/2} \sqrt{\hat{p}\hat{q}/n} \leq \hat{p} - p \leq z_{\alpha/2} \sqrt{\hat{p}\hat{q}/n}\right) = 1 - \alpha.$$

- Further simplifying (multiply by -1 and add $\hat{p}$), we obtain

$$\mathbb{P}\left(\hat{p} - z_{\alpha/2} \sqrt{\hat{p}\hat{q}/n} \leq p \leq \hat{p} + z_{\alpha/2} \sqrt{\hat{p}\hat{q}/n}\right) = 1 - \alpha.$$

- Thus, the $100(1 - \alpha) \%$ confidence interval for p is given by:

$$\left[\hat{p} - z_{\alpha/2} \sqrt{\hat{p}\hat{q}/n}, \hat{p} + z_{\alpha/2} \sqrt{\hat{p}\hat{q}/n}\right], \text{ or, simply, } \hat{p} \pm z_{\alpha/2} \sqrt{\hat{p}\hat{q}/n}.$$

Compare with μ : replace $\bar{X}_n$ by $\hat{p}$ and S by $\sqrt{\hat{p}\hat{q}}$ (the SD of 0/1 data, up to $\sqrt{n/(n-1)}$).

ESTIMATING MEANS AND PROPORTIONS



- For a quantitative population,

Point estimator of population mean $\mu : \bar{x}$

Margin of error ($n \geq 30$) : $\pm 1.96 \frac{s}{\sqrt{n}}$

- For a binomial population,

Point estimator of population proportion $p : \hat{p} = x/n$

Margin of error ($n \geq 30$) : $\pm 1.96 \sqrt{\frac{\hat{p}\hat{q}}{n}}$

EXAMPLE



- A homeowner randomly samples 64 homes similar to her own and finds that the average selling price is \$252,000 with a standard deviation of \$15,000. Estimate the average selling price for all similar homes in the city.

Point estimator of μ : $\bar{x} = 252,000$

$$\text{Margin of error : } \pm 1.96 \frac{s}{\sqrt{n}} = \pm 1.96 \frac{15,000}{\sqrt{64}} = \pm 3675$$

EXAMPLE



A quality control technician wants to estimate the proportion of soda cans that are underfilled.

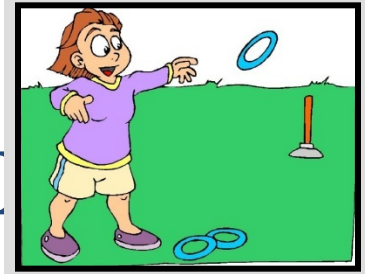
He randomly samples 200 cans of soda and finds 10 underfilled cans.

$n = 200$ $p = \text{proportion of underfilled cans}$

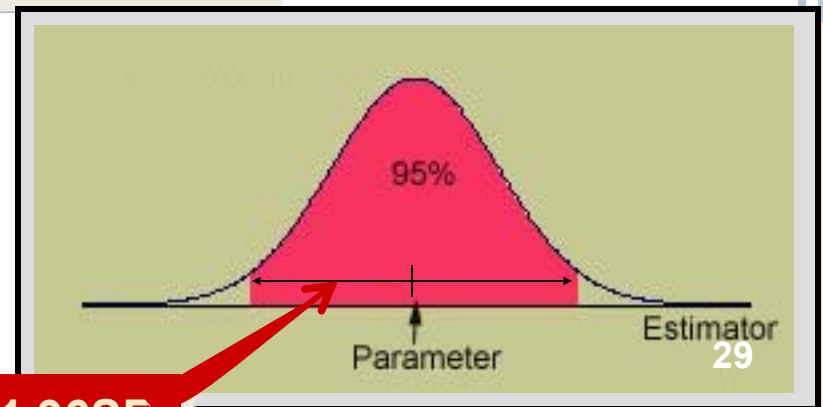
Point estimator of p : $\hat{p} = x/n = 10/200 = .05$

Margin of error : $\pm 1.96 \sqrt{\frac{\hat{p}\hat{q}}{n}} = \pm 1.96 \sqrt{\frac{(.05)(.95)}{200}} = \pm .03$

INTERVAL ESTIMATION

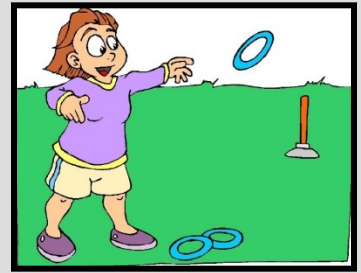


- Create an interval (a, b) so that you are fairly sure that the parameter lies between these two values.
- “Fairly sure” means “with high probability”, measured using the **confidence coefficient**, $1-\alpha$.
Usually, $1-\alpha = .90, .95, .98, .99$
- Suppose $1-\alpha = .95$ and that the estimator has a normal distribution.

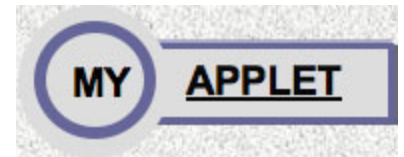
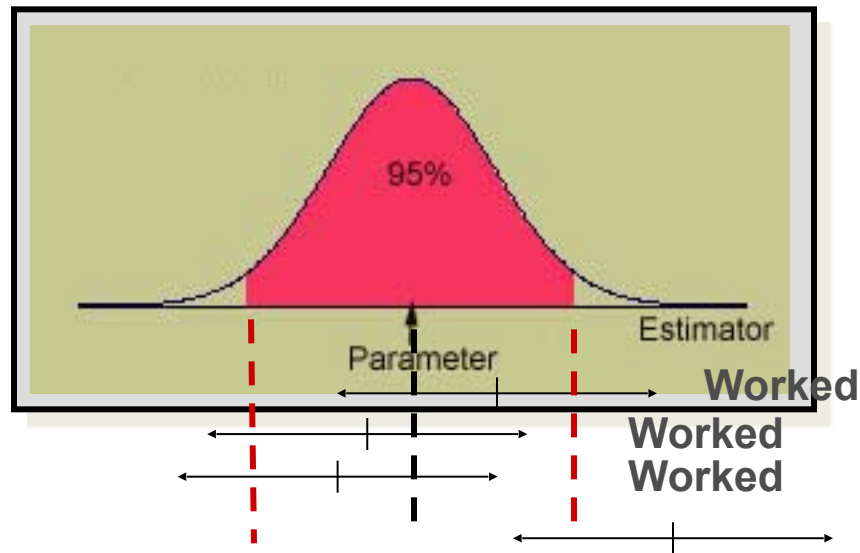


Parameter $\pm 1.96SE$

INTERVAL ESTIMATION



- Since we don't know the value of the parameter, consider **Estimator $\pm 1.96SE$** which has a variable center.

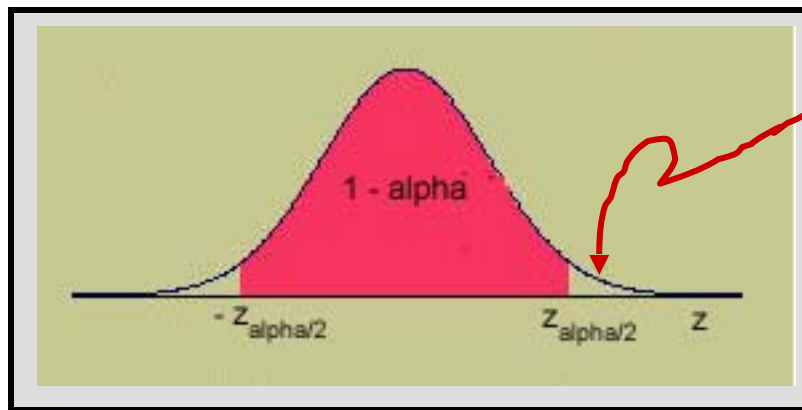


- Only if the estimator falls in the tail areas will the interval fail to enclose the parameter. This happens only 5% of the time.

TO CHANGE THE CONFIDENCE LEVEL



- To change to a general confidence level, $1-\alpha$, pick a value of z that puts area $1-\alpha$ in the center of the z distribution.



Tail area	$z_{\alpha/2}$
.05	1.645
.025	1.96
.01	2.33
.005	2.58

100(1- α)% Confidence Interval:

$$\text{Estimator} \pm z_{\alpha/2} \text{SE}$$

CONFIDENCE INTERVALS FOR MEANS AND PROPORTIONS



- For a quantitative population,

Confidence interval for a population mean μ :

$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

- For a binomial population,

Confidence interval for a population proportion p :

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

EXAMPLE



- A random sample of $n = 50$ males showed a mean average daily intake of dairy products equal to 756 grams with a standard deviation of 35 grams. Find a 95% confidence interval for the population average μ .

$$\bar{x} \pm 1.96 \frac{s}{\sqrt{n}} \Rightarrow 756 \pm 1.96 \frac{35}{\sqrt{50}} \Rightarrow 756 \pm 9.70$$

or $746.30 < \mu < 765.70$ grams.

EXAMPLE



- Find a 99% confidence interval for μ , the population average daily intake of dairy products for men.

$$\bar{x} \pm 2.58 \frac{s}{\sqrt{n}} \Rightarrow 756 \pm 2.58 \frac{35}{\sqrt{50}} \Rightarrow 756 \pm 12.77$$

or $743.23 < \mu < 768.77$ grams.

The interval must be wider to provide for the increased confidence that it does indeed enclose the true value of μ .



EXAMPLE

- Of a random sample of $n = 150$ college students, 104 of the students said that they had played on a soccer team during their K-12 years. Estimate the proportion of college students who played soccer in their youth with a 98% confidence interval.

$$\hat{p} \pm 2.33 \sqrt{\frac{\hat{p}\hat{q}}{n}} \Rightarrow \frac{104}{150} \pm 2.33 \sqrt{\frac{.69(.31)}{150}}$$
$$\Rightarrow .69 \pm .09 \quad \text{or } .60 < p < .78.$$



8.4 ESTIMATING THE DIFFERENCE BETWEEN TWO POPULATION AND MEANS (SKIPPED 20251122)

ESTIMATING THE DIFFERENCE BETWEEN TWO MEANS

- Sometimes we are interested in comparing the means of two populations.
 - The average growth of plants fed using two different nutrients.
 - The average scores for students taught with two different teaching methods.
- To make this comparison,

A random sample of size n_1 drawn from population 1 with mean μ_1 and variance σ_1^2 .

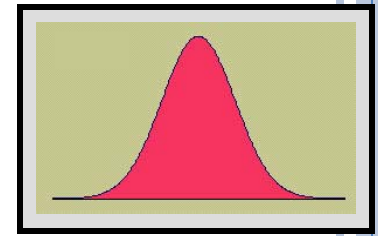
A random sample of size n_2 drawn from population 2 with mean μ_2 and variance σ_2^2 .

ESTIMATING THE DIFFERENCE BETWEEN TWO MEANS

- We compare the two averages by making inferences about $\mu_1 - \mu_2$, the difference in the two population averages.
 - If the two population averages are the same, then $\mu_1 - \mu_2 = 0$.
 - The best estimate of $\mu_1 - \mu_2$ is the difference in the two sample means,

$$\bar{x}_1 - \bar{x}_2$$

THE SAMPLING DISTRIBUTION OF $\bar{x}_1 - \bar{x}_2$



1. The mean of $\bar{x}_1 - \bar{x}_2$ is $\mu_1 - \mu_2$, the difference in the population means.

2. The standard deviation of $\bar{x}_1 - \bar{x}_2$ is $SE = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$.

3. If the sample sizes are large, the sampling distribution of $\bar{x}_1 - \bar{x}_2$ is approximately normal, and SE can be estimated

as $SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$.

Key ingredients

1. Model

- $X_{1i} \sim^{i.i.d.} \mathcal{G}_1(\mu_1, \sigma_1^2)$ for $i = 1, \dots, n_1$
- $X_{2j} \sim^{i.i.d.} \mathcal{G}_2(\mu_2, \sigma_2^2)$ for $j = 1, \dots, n_2$

2. We are interested in understanding $(\mu_1 - \mu_2)$.

3. Point estimation: $(\bar{X}_1 - \bar{X}_2)$, where

$$\bar{X}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} X_{1i}, \quad \bar{X}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} X_{2j}.$$

- Its mean is $(\mu_1 - \mu_2)$.

- The standard error is $SE = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

CLT

- With CLT and math, we have

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \xrightarrow{d} Z$$

THE SLUSKY'S THEOREM (ADVANCED CLT)

- When σ_1 and σ_2 are unknown, they are estimated by

- $SE = \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$, where

-

$$S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_{1i} - \bar{X}_1)^2, \quad S_2^2 = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (X_{2j} - \bar{X}_2)^2.$$

- When the sample sizes are large, the advanced CLT

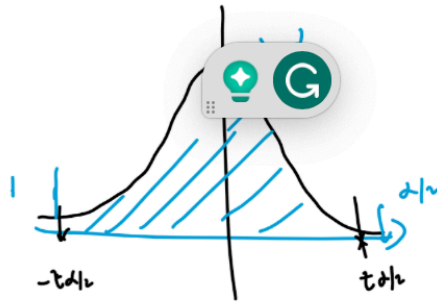
gives
$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \xrightarrow{d} Z.$$

CI FOR $(\mu_1 - \mu_2)$

Fig 15

$$\frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \rightarrow Z$$

$$1-\alpha = P\left(-z_{\alpha/2} < \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} < z_{\alpha/2}\right)$$



$$= P\left(-z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < (\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2) < z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}\right)$$

$$= P\left(-z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < (\mu_1 - \mu_2) - (\bar{x}_1 - \bar{x}_2) < z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}\right)$$

$$= P\left((\bar{x}_1 - \bar{x}_2) - z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < (\mu_1 - \mu_2) < (\bar{x}_1 - \bar{x}_2) + z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}\right)$$

Then $100(1-\alpha)\%$'s C.I for

$(\mu_1 - \mu_2)$ is $\left[(\bar{x}_1 - \bar{x}_2) - z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, (\bar{x}_1 - \bar{x}_2) + z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \right]$

Textbook p.309

● Need a Tip?

Right Tail Area	z-Value
.05	1.645
.025	1.96
.01	2.33
.005	2.58

Large-Sample Point Estimation of $(\mu_1 - \mu_2)$

Point estimator: $(\bar{x}_1 - \bar{x}_2)$

$$95\% \text{ Margin of error: } \pm 1.96 \text{ SE} = \pm 1.96 \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

A $(1 - \alpha)100\%$ Large-Sample Confidence Interval for $(\mu_1 - \mu_2)$

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

How do we interpret this CI?

$$1 - \alpha = P\left(\underbrace{(\bar{x}_1 - \bar{x}_2) - z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}_{\text{lower bound}} < \underbrace{(\mu_1 - \mu_2)}_{\text{true parameter}} < \underbrace{(\bar{x}_1 - \bar{x}_2) + z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}_{\text{upper bound}}\right).$$

In repeated sampling, about $100(1 - \alpha)\%$ intervals constructed this way will cover the true parameter $(\mu_1 - \mu_2)$.

Textbook: p. 311

Statistic	Normal—All Sample Sizes	Nonnormal—Large Samples
$\frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	z-statistic	Approximate z-statistic
$\frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$	See Chapter 10	Approximate z-statistic

EXAMPLE



Avg Daily Intakes	Men	Women
Sample size	50	50
Sample mean	756	762
Sample Std Dev	35	30

- Compare the average daily intake of dairy products of men and women using a 95% confidence interval.

$$\begin{aligned} & (\bar{x}_1 - \bar{x}_2) \pm 1.96 \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \\ \Rightarrow & (756 - 762) \pm 1.96 \sqrt{\frac{35^2}{50} + \frac{30^2}{50}} \Rightarrow -6 \pm 12.78 \\ \text{or } & -18.78 < \mu_1 - \mu_2 < 6.78. \end{aligned}$$

EXAMPLE, CONTINUED



$$-18.78 < \mu_1 - \mu_2 < 6.78$$

- Could you conclude, based on this confidence interval, that there is a difference in the average daily intake of dairy products for men and women?
- The confidence interval contains the value $\mu_1 - \mu_2 = 0$. Therefore, it is possible that $\mu_1 = \mu_2$. You would not want to conclude that there is a difference in average daily intake of dairy products for men and women.

A BETTER WAY (BUT RELATED TO H_0)

- Assume $\mu_1 = \mu_2$.
- CI covers 0 $\Leftrightarrow Z^*$ does not fall into the rejection region.
- CI does not cover 0 equals $\Leftrightarrow Z^*$ fall into the rejection region.



8.5 ESTIMATING THE DIFFERENCE BETWEEN TWO BINOMIAL PROPORTIONS

SKIPPED

ESTIMATING THE DIFFERENCE BETWEEN TWO PROPORTIONS

- Sometimes we are interested in comparing the proportion of “successes” in two binomial populations.
 - The germination rates of untreated seeds and seeds treated with a fungicide.
 - The proportion of male and female voters who favor a particular candidate for governor.

A random sample of size n_1 drawn from binomial population 1 with parameter p_1 .

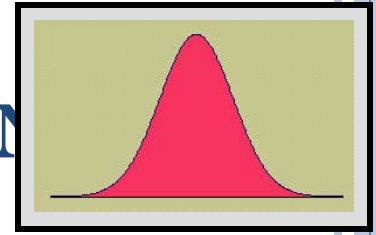
A random sample of size n_2 drawn from binomial population 2 with parameter p_2 .

ESTIMATING THE DIFFERENCE BETWEEN TWO MEANS

- We compare the two proportions by making inferences about $p_1 - p_2$, the difference in the two population proportions.
 - If the two population proportions are the same, then $p_1 - p_2 = 0$.
 - The best estimate of $p_1 - p_2$ is the difference in the two sample proportions,

$$\hat{p}_1 - \hat{p}_2 = \frac{x_1}{n_1} - \frac{x_2}{n_2}$$

THE SAMPLING DISTRIBUTION OF



1. The mean of $\hat{p}_1 - \hat{p}_2$ is $p_1 - p_2$, the difference in the population proportions.

2. The standard deviation of $\hat{p}_1 - \hat{p}_2$ is $SE = \sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}$.

3. If the sample sizes are large, the sampling distribution of $\hat{p}_1 - \hat{p}_2$ is approximately normal, and SE can be estimated

as $SE = \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}$.



ESTIMATING $P_1 - P_2$



- For large samples, point estimates and their margin of error as well as confidence intervals are based on the standard normal (z) distribution.

Point estimate for $p_1 - p_2 : \hat{p}_1 - \hat{p}_2$

Margin of Error : $\pm 1.96 \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}$

Confidence interval for $p_1 - p_2 :$

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}$$

KEY INGREDIENTS

1. Model

- $X_{1i} \sim^{i.i.d.} \text{Bernoulli}(p_1)$ for $i = 1, \dots, n_1$
- $X_{2j} \sim^{i.i.d.} \text{Bernoulli}(p_2)$ for $j = 1, \dots, n_2$

2. We are interested in understanding $(p_1 - p_2)$.

3. Point estimator: $(\hat{p}_1 - \hat{p}_2)$ where

$$\hat{p}_1 = \frac{\sum_{i=1}^{n_1} X_{1i}}{n_1}, \quad \hat{p}_2 = \frac{\sum_{j=1}^{n_2} X_{2j}}{n_2}.$$

KEY INGREDIENTS

- The mean of $(\hat{p}_1 - \hat{p}_2)$ is $(p_1 - p_2)$
- The standard error of $(\hat{p}_1 - \hat{p}_2)$ is by

$$\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}, \text{ and is estimated by}$$

$$\sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}$$

CLT: $\frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}} \xrightarrow{d} Z$

IN-CLASS EXERCISE

- Derive the $(1 - \alpha) \%$ confidence interval for $p_1 - p_2$.

TEXTBOOK: p. 314

Large-Sample Point Estimation of $(p_1 - p_2)$

Point estimator: $(\hat{p}_1 - \hat{p}_2)$

$$\text{95\% Margin of error} \pm 1.96 \text{ SE} = \pm 1.96 \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}$$

Textbook: p.315

A $(1 - \alpha)100\%$ Large-Sample Confidence Interval for $(p_1 - p_2)$

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}} \quad 100(1 - \alpha) \% \text{ Margin of error}$$

Assumption: n_1 and n_2 must be sufficiently large so that the sampling distribution of $(\hat{p}_1 - \hat{p}_2)$ can be approximated by a normal distribution—namely, if $n_1 \hat{p}_1$, $n_1 \hat{q}_1$, $n_2 \hat{p}_2$, and $n_2 \hat{q}_2$ are all greater than 5.

EXAMPLE



Youth Soccer	Male	Female
Sample size	80	70
Played soccer	65	39

- Compare the proportion of male and female college students who said that they had played on a soccer team during their K-12 years using a 99% confidence interval.

$$(\hat{p}_1 - \hat{p}_2) \pm 2.58 \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}$$
$$\Rightarrow \left(\frac{65}{80} - \frac{39}{70} \right) \pm 2.58 \sqrt{\frac{.81(.19)}{80} + \frac{.56(.44)}{70}} \Rightarrow .25 \pm .19$$

or $.06 < p_1 - p_2 < .44$.

EXAMPLE, CONTINUED

$$.06 < p_1 - p_2 < .44$$



- Could you conclude, based on this confidence interval, that there is a difference in the proportion of male and female college students who said that they had played on a soccer team during their K-12 years?
- The confidence interval does not contain the value $p_1 - p_2 = 0$. Therefore, it is not likely that $p_1 = p_2$. You would conclude that there is a difference in the proportions for males and females.

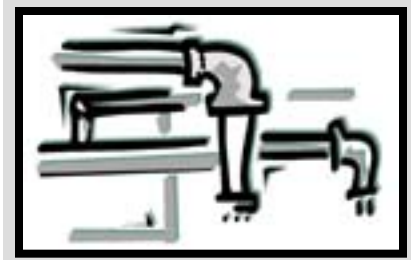
A higher proportion of males than females played soccer in their youth.

The slide features a dark blue background. On the left side, there are several vertical stripes of varying shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. The largest circle is positioned near the top left, and several smaller circles are arranged vertically below it, some overlapping the stripes.

8.7 CHOOSING THE SAMPLE SIZE

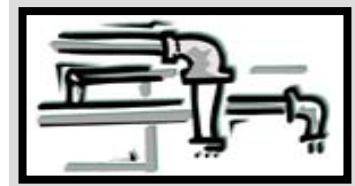
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CHOOSING THE SAMPLE SIZE



- The total amount of relevant information in a sample is controlled by two factors:
 - The **sampling plan** or **experimental design**: the procedure for collecting the information
 - The **sample size n** : the amount of information you collect.
- In a statistical estimation problem, the accuracy of the estimation is measured by the **margin of error** or the **width of the confidence interval**.

Choosing the Sample Size



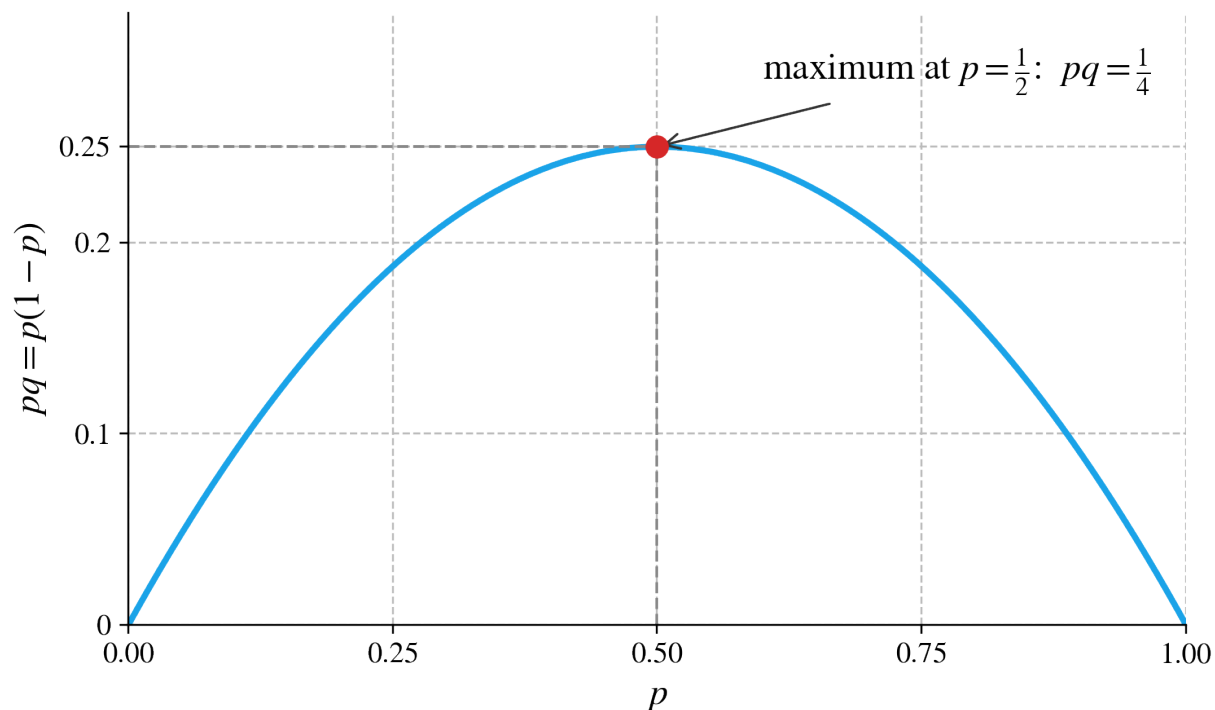
1. Determine the size of the **margin of error**, B , that you are willing to tolerate.
2. Choose the sample size by solving for n in the inequality: $1.96 \text{ SE} \leq B$, where SE is a function of the sample size n .

HOW TO FIND n ?

$$1.96\sqrt{\frac{\sigma^2}{n}} \leq B \iff \frac{\sigma^2}{n} \leq \frac{B^2}{1.96^2} \iff n \geq \frac{1.96^2 \sigma^2}{B^2}$$

1. For quantitative populations, estimate the population standard deviation using a previously calculated value of **s** or the range approximation $\sigma \approx \mathbf{Range} / 4$.
2. For binomial populations, use the conservative approach and approximate p using the value **$p = .5$** .

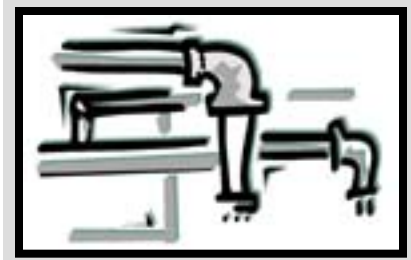
WHY $p = 1/2$ IS CONSERVATIVE?



- General confidence coefficient

$$z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq B \iff n \geq \frac{z_{\alpha/2}^2 \sigma^2}{B^2}.$$

EXAMPLE



A producer of PVC pipe wants to survey wholesalers who buy his product in order to estimate the proportion who plan to increase their purchases next year. What sample size is required if he wants his estimate to be within .04 of the actual proportion with probability equal to .95?

$$\begin{aligned} 1.96\sqrt{\frac{pq}{n}} &\leq .04 &\Rightarrow 1.96\sqrt{\frac{.5(.5)}{n}} &\leq .04 \\ \Rightarrow \sqrt{n} &\geq \frac{1.96\sqrt{.5(.5)}}{.04} = 24.5 &\Rightarrow n &\geq 24.5^2 = 600.25 \end{aligned}$$

He should survey at least 601 wholesalers.

例子：選舉民調要多少有效樣本？

民調中心要估計台北市長候選人甲的支持率 p 。要求：在 95% 信心水準下，抽樣誤差(i.e., 95% margin of error, 誤差範圍) 不超過 ± 3 個百分點。至少要訪問多少位有效樣本？

Sol. 

1. 可容忍的誤差 $B = 0.03$ 。
2. 解 $1.96 \cdot SE \leq B$ ，其中 $SE = \sqrt{(pq/n)}$ 。
3. 選前不知道 p ，用保守值 $p = 0.5$ ($pq = 0.25$ 最大，算出的 n 最大)。
4. $1.96 \cdot \sqrt{(0.25/n)} \leq 0.03 \Rightarrow \sqrt{n} \geq 1.96 \times 0.5 / 0.03 = 32.67 \Rightarrow n \geq 32.67^2 = 1067.1$ 。

Answer: 至少 1,068 位有效樣本。

這就是為什麼台灣民調的有效樣本幾乎都在 1,000 出頭。

反過來算（新聞的寫法）

- $n = 1,120$ 時，誤差 $= 1.96 \times \sqrt{(0.25/1120)} = 0.0293$ ，所以新聞寫「抽樣誤差為 ± 2.9 個百分點」。

台北市長選戰最新（中時新聞網 2026-10-08）

本次民調是TVBS民意調查中心115年10月3日、5日至7日晚間18:30至22:00所進行調查，共接觸1,284位20歲以上台北市民，其中拒訪為164位，拒訪率為12.8%，最後成功訪問有效樣本1,120位，在95%的信心水準下，抽樣誤差為 ± 2.9 個百分點以內。抽樣方法採用市內電話號碼後四碼隨機抽樣，人員電話訪問，所有資料並依母體性別、年齡、地區、教育程度等變項進行統計加權處理，調查經費來源為TVBS。

<https://www.chinatimes.com/realtimenews/20261008004675-260407?chdtv>

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CHAPTER SUMMARY

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KEY CONCEPTS

I. Types of Estimators

1. **Point estimator:** a single number is calculated to estimate the population parameter.
2. **Interval estimator:** two numbers are calculated to form an interval that contains the parameter.

II. Properties of Good Estimators

1. **Unbiased:** the average value of the estimator equals the parameter to be estimated.
2. **Minimum variance:** of all the unbiased estimators, the best estimator has a sampling distribution with the smallest standard error.
3. The **margin of error** measures the maximum distance between the estimator and the true value of the parameter.

KEY CONCEPTS

III. Large-Sample Point Estimators

To estimate one of four population parameters when the sample sizes are large, use the following point estimators with the appropriate margins of error.

Parameter	Point Estimator	Margin of Error
μ	$\bar{x}$	$\pm 1.96 \left(\frac{s}{\sqrt{n}} \right)$
p	$\hat{p} = \frac{x}{n}$	$\pm 1.96 \sqrt{\frac{\hat{p}\hat{q}}{n}}$
$\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$	$\pm 1.96 \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
$p_1 - p_2$	$(\hat{p}_1 - \hat{p}_2) = \left(\frac{x_1}{n_1} - \frac{x_2}{n_2} \right)$	$\pm 1.96 \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$

KEY CONCEPTS

IV. Large-Sample Interval Estimators

To estimate one of four population parameters when the sample sizes are large, use the following interval estimators.

Parameter	$(1 - \alpha)100\%$ Confidence Interval
μ	$\bar{x} \pm z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right)$
p	$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$
$\mu_1 - \mu_2$	$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
$p_1 - p_2$	$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$

KEY CONCEPTS

1. All values in the interval are possible values for the unknown population parameter.
2. Any values outside the interval are unlikely to be the value of the unknown parameter.
3. To compare two population means or proportions, look for the value 0 in the confidence interval. If 0 is in the interval, it is possible that the two population means or proportions are equal, and you should not declare a difference. If 0 is not in the interval, it is unlikely that the two means or proportions are equal, and you can confidently declare a difference.

V. One-Sided Confidence Bounds

Use either the upper (+) or lower (-) two-sided bound, with the critical value of z changed from $z_{\alpha/2}$ to z_{α} .